

S**PAYTM SAATHI / BUILD STATUS AND DEMO GUIDE**

What we have built, what is on each screen, and what to ask

Track 2: AI-Powered Financial Journeys. An event-first financial resolution layer: the customer says what happened, Saathi gathers source-linked evidence, calculates the exact funding gap, compares safe options, and coordinates the approved step through partners until it is resolved.

Tell once. See the evidence. Choose the next step.

Status as of 3 Oct 2026

Commit c25e85d on main (pushed)

Team Swamit Labs

Synthetic demo data only

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01 At a glance

A working, end-to-end prototype: React web app, Node.js + TypeScript API, a LangGraph.js agent graph, YAML playbooks, and real MCP partner servers behind a trust gateway.

5

playbooks (journeys), one engine

7

partner MCP servers

30

MCP tools, 10 approval-gated writes

12

agent nodes on LangGraph.js

11

languages: voice, replies, read-aloud

60

automated tests, all passing

Journeys that work end to end

- Hospital bill with an insurance gap (hero)
- Unrecognised UPI debit (fraud dispute)
- Failed UPI payment, refund not received
- EMI shortfall before payday
- Anything else: specialist handoff

Live AI services (keys in .env)

- OpenAI: follow-up Q&A on a case, evidence summary, photo OCR
- Sarvam: speech-to-text, translation, text-to-speech
- All gated by per-request consent

The rule that never bends

- AI understands, explains, and prepares
- Deterministic code does every rupee of maths
- The customer approves the exact payload
- Partners make claim, credit, and dispute decisions

02 Architecture as built

This follows the deck's layers. Where we deliberately differ from the plan document (FastAPI, Postgres, Redis), the build uses Node.js, SQLite, and in-process events so the demo runs on one laptop with one command.

EXPERIENCE

React + TypeScript (Vite). Home (Financial Twin), Saathi chat with mic and language picker, case workspace with Plan, Evidence, Agents, Timeline, Passport, and Trust ledger tabs, specialist desk. Live updates over Server-Sent Events. Phone layout.

API EDGE

Express 5. Demo JWT login with role scopes, case ownership on every read, login rate limiting, CSP and security headers, signed partner callbacks, idempotency keys.

AI ORCHESTRATION

LangGraph.js state graph: classifier, consent gate, context retriever, policy RAG analyst, bill / transaction / EMI auditor, decision service, resolution explainer, human-review branch, plus action preparer and tracker. Every run is traced (nodes, tools, timings).

PLAYBOOKS

`data/playbooks/*.yaml` : triggers (English, Hinglish, Devanagari), urgency, rules engine, allowed tools, exit. Failed UPI refund is fully declarative (rules written as safe expressions in YAML).

DECISION SERVICES

Coverage rules engine (line-by-line), exact-gap and EMI-shortfall calculators, EMI and APR maths, affordability guardrail, emergency-buffer guardrail, confidence gate, option scoring on cost, risk, time, and effort. Commission is never an input.

MCP GATEWAY

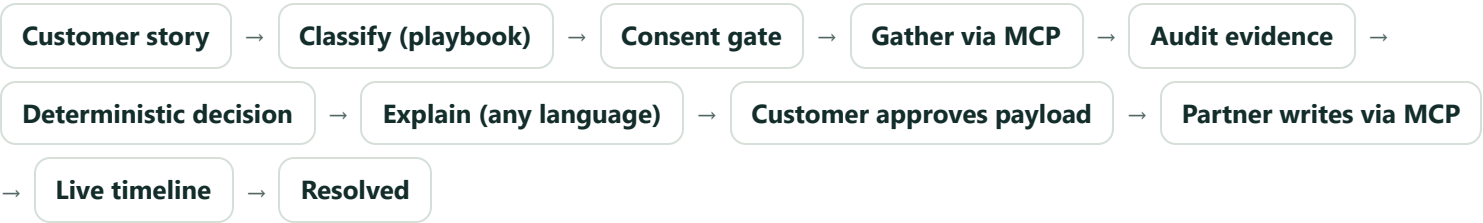
Allowlist, strict schemas, case scope, playbook scope, consent, approval token bound to case + action + person + payload hash, write de-duplication, audit of every allow and deny. Then a real MCP client call.

PARTNER MCP SERVERS

`insurer.v1` , `provider.v1` (hospital), `payments.v1` , `lender.v1` , `aa.v1` (Account Aggregator), `crm.v1` (Paytm Support), `knowledge.v1` . In-process by default; `MCP_URL_<SERVER>` switches any of them to a remote Streamable HTTP server. Also exposed at `/mcp/<server>` .

DATA

SQLite (cases, audit events, preferences, partner events). Synthetic fixtures in `data/` . Original uploads are never stored; only redacted text.



03 What we built, step by step

STEP	WHAT IT ADDED	COMMIT
Base (team)	React + Express rewrite: demo JWT login, case API, MCP-style gateway, evidence and uploads, deterministic hospital and UPI decisions, approval-bound actions, simulated partner, Passport, Trust ledger, support queue.	7c7dbf0
Agent graph	Workflow moved onto a LangGraph.js state graph with traced runs and an Agents tab; Hinglish replies built only from decision numbers.	f274972
EMI journey	EMI auditor node, loan and salary data, shortfall maths; recommends moving the due date past payday over borrowing.	a8388c5
Hardening	Login rate limit, CSP and frame-blocking headers, graceful shutdown, Dockerfile, GitHub Actions CI.	37f0905
Voice + languages + AI chat	Mic (Sarvam STT), 11-language picker, translated replies, read-aloud, OpenAI follow-up Q&A with a numeric guard.	550f5f6
Live updates, OCR, demo command	Server-Sent Events, photo OCR via OpenAI vision (consented), <code>npm run demo</code> , watch fix.	746b905
Complete product loop	Proactive alerts, specialist desk, customer-confirmed bill parsing, Passport PDF, phone layout, <code>npm run reset</code> .	f6a124c
Live provider fixes	Translation per sentence with amount checks, current Sarvam voice model, native-script answers, Devanagari classification.	a194383
Financial Twin + Home	Sourced net position, cash flow, bills inbox with projected balance, money tiles, goals; Home tab with mode chips.	5de245f
Playbook registry	All journeys as YAML; gateway denies undeclared tools; Failed UPI refund added as config (RBI T+1 and INR 100/day compensation).	fc10dbf
Real MCP servers	Seven SDK-built partner servers, gateway as MCP client, HTTP endpoints for MCP Inspector; payment link, Key Fact Statement, Account Aggregator, CRM handoff tools.	ec3ba30
Coverage rules engine	Cover calculated per bill line (room cap, non-medical exclusion, deductible, sum insured) with clause and reason for each rupee.	c25e85d

04 Plan, deck, and idea coverage

The idea document's 10 core hackathon capabilities

CAPABILITY	STATUS	WHERE TO SEE IT
Conversational home: voice, chat, Hinglish	DONE	Home tab + Saathi tab (mic, 11 languages)
Mini Financial Twin	DONE	Home tab
Hospital bill upload	DONE	Evidence tab: TXT, PDF, or photo (OCR); confirm total in Plan
Insurance policy upload	PARTIAL	Accepted and cited; a specialist verifies it before automated steps
Policy RAG with citations	DONE	Evidence tab (keyword index, not vectors)
Exact-gap calculator	DONE	Plan tab gap tiles + coverage breakdown
Resolution graph	DONE	Plan tab option comparison with "Why this score?"
Affordability engine	DONE	Guardrails on every credit option
Consent + one-click approved action	DONE	Consent checkbox, Review & approve with payload hash
Live resolution timeline	DONE	Timeline tab (Server-Sent Events)

Architecture deck claims

DECK CLAIM	STATUS	NOTES
7 agents on LangGraph	DONE	LangGraph.js; node names differ slightly from the slide
A new problem is a new YAML playbook	DONE	Failed UPI refund proves it
MCP-native partners, standard contracts, swap by URL	DONE	Proven live with remote insurer and lender servers
Coverage rules: payable / capped / excluded per line	DONE	coverage-v1
Key Fact Statement before credit	DONE	Bound into the approval hash
Account Aggregator consent and data	DONE	Simulated AA; consent handle shown in Agents tab
Consent with ID and 72h expiry; hash-chained audit ledger	NOT YET	Purpose-bound, revocable consent exists; no expiry or hash chain yet
Circuit breaker and response sanitiser in gateway	NOT YET	Partner failures already route to a specialist
Postgres, pgvector, Redis, object storage	SWAPPED	SQLite and in-process events for the demo; say so when presenting
Golden demo scenarios, latency budget display	PARTIAL	60 tests; per-node timings in Agents tab

05 The UI, screen by screen

Open `http://localhost:5173` (development) or `http://127.0.0.1:8000` (after `npm run demo`).

1. Login

- Three synthetic accounts: Riya (hero), Arjun (healthy savings), Support agent
- Banner: synthetic data, simulated partner outcomes
- Top bar shows API, OpenAI, Sarvam, and partner-channel status

2. Home tab: the Financial Twin

- Greeting and status headline, for example "INR 41,500 is due before your salary on 10 Oct, but your balance is INR 23,400"
- Net financial position (assets minus liabilities)
- **Ask Saathi** box and mode chips: RECOVER, PROTECT, PLAN
- **Needs attention:** shortfall before payday, renewals, Saathi noticed alerts
- **Financial inbox:** next 30 days of bills and salary, projected balance after each
- Money tiles, goals with progress, monthly cash flow

3. Saathi tab: the conversation

- Language picker (Auto, English, Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia)
- Demo buttons: Hospital, Unrecognized UPI, Failed UPI refund, EMI shortfall
- Alerts opt-in and Saathi noticed cards
- Consent box: "Allow Saathi to read my case records"
- Mic button (Sarvam), with its own voice consent
- Speaker button on every Saathi reply; "Original (English)" under translations
- With a case open: **Ask Saathi** follow-up questions (OpenAI, own consent)

4. Case panel: header and status

- Case ID, status, urgency, event type
- Stepper: Intake, Understand, Evidence ready, Options ready, Awaiting approval, In progress, Resolved (or Human review)
- "Talk to a human" button; consent banner if not yet granted

5. Plan tab

- Gap tiles: bill minus cover minus what you pay = exact gap
- **How the cover was calculated:** each bill line, status, clause, reason, deductible
- Options ranked with score bars, guardrails, trade-offs; "Why this score?"
- **Review & approve:** exact steps, partners, amounts, Key Fact Statement for credit, payload hash
- Transaction picker (UPI / refund) and bill-total confirmation (uploads)

6. Evidence tab

- Facts table: value, source, confidence, assumptions
- Cited policy clauses or playbook steps
- Missing documents warning
- Upload a bill or policy (TXT, PDF, JPG/PNG with OCR consent)
- OpenAI evidence summary (consented)

7. Agents tab

- Active playbook card: version, engine, allowed read tools, write tools highlighted
- Each graph run: trigger, nodes in order, MCP tools each node called, timing, outcome
- Shows Account Aggregator consent handle and money authority per node

8. Timeline, Passport, Trust ledger

- **Timeline:** every step, live partner callbacks
- **Passport:** Tell-it-once packet; Download PDF, Copy summary, JSON
- **Trust ledger:** every gateway allow and deny, approvals, partner events; tool catalogue

9. Specialist desk (support login)

- Queue, human-review cases first
- Pick up the case, message the customer or save an internal note
- Verify customer documents and recalculate
- Recommend an option (the customer still approves), close the case

10. Behind the scenes, worth showing

- `GET /api/playbooks` , `/api/mcp/servers` , `/api/agent/graph`
- MCP Inspector on `/mcp/insurer` and other servers
- Live timeline updates without refresh

06 What to ask on the UI

Type these in the Saathi tab (or use the demo buttons and Home chips). Tick **Allow Saathi to read my case records** first, or the case pauses at the consent gate (that pause is itself a good thing to show). Expected results are for Riya on fresh data.

Hospital bill (hero journey)

`Papa hospital mein hain. Bill INR 80,000 hai. Insurance hai, ab kya karun?`

Hospitalization, high urgency, reply in Hinglish. Cover INR 55,000 (room capped at INR 15,000, INR 5,000 consumables excluded, INR 5,000 deductible). $INR\ 80,000 - 55,000 - 10,000 = \text{INR } 15,000 \text{ gap}$. Best option: **Claim + 3-month plan (91.4)**, then specialist (83.8), savings (69.3), wait (57.5), full loan (44.3). Approval shows claim, document request, INR 10,000 payment link, INR 15,000 loan with its Key Fact Statement. Partner updates arrive until **Resolved**.

Also try:

My father is admitted in hospital and the bill is INR 80,000. What should I do?

मेरे पापा अस्पताल में हैं। बिल 80,000 रुपये है।

English reply for the first; the Devanagari message is still recognised as a hospital case. Pick Hindi or Tamil in the language picker to get the explanation translated (amounts unchanged).

Follow-up questions on an open case (OpenAI, tick "Allow Saathi AI")

Why is the 3-month plan better than using my savings?

Mujhe kitna loan lena padega?

What documents are still missing for the claim?

What happens if I don't approve anything?

Why is my room charge not fully covered?

Answers use only this case's facts and numbers (for example INR 15,000 gap, INR 501 cost, INR 3,000 left in savings). If the model invents a number, Saathi replaces the answer with its own explanation and logs it in the Trust ledger.

Unrecognised UPI debit (fraud)

Mere account se INR 8,500 ka UPI payment hua jo maine nahi kiya. Kya karun?

Two INR 8,500 debits listed. Pick the 2:14 AM QuickKart one, "I don't recognize this": signals are unrecognised device, first-time payee, unusual hour. Best option: **Open a dispute + secure the account (95.5)**. After approval, provisional credit arrives on the timeline. Choosing "This was me" closes the case with nothing filed.

Failed UPI payment, refund missing (declarative YAML playbook)

INR 2,450 ka UPI payment failed ho gaya, paise kat gaye par refund nahi aaya.

Picker shows the failed INR 2,450 electricity payment ("This is the payment"). Facts: 3 days since debit, T+1 deadline, **2 days late, INR 200 compensation**. Best option: **Raise a refund trace and claim compensation (95.8)**. Agents tab shows playbook `failed_upi_refund` as "Declarative YAML".

EMI shortfall before payday

Salary delayed hai, is mahine EMI bharne ke paise kam hain. Kya options hain?

INR 12,000 EMI due 5 Oct, INR 7,400 available, **INR 4,600 short**, salary on 10 Oct. Best option: **Move the EMI date past payday (85.3)** for a INR 250 fee; then specialist, a INR 4,600 bridge loan, savings (below buffer), and letting it bounce (most expensive).

Anything else

Help me plan my monthly money.

No playbook matches, so a specialist route with a CRM handoff ticket is offered.

Upload your own bill (Evidence tab)

Start a hospital case, then upload this as a .txt file (or a photo of a printed bill with OCR consent):

```
City Care Hospital – Final Bill
Room charges (3 days)  36,000
Doctor visits  12,500
Medicines & consumables  18,250
Lab tests  8,250
Grand Total  75,000
```

Saathi reads 4 lines and INR 75,000 and asks you to confirm. After "Yes, this is correct": room capped at INR 15,000, cover INR 49,000, **gap INR 16,000**, with a note that consumables were not itemised. Uploading a policy instead pauses automated steps until a specialist verifies it.

Other things to try

TRY	WHAT YOU SHOULD SEE
Sign in as Arjun (1357), run the hospital prompt	Recommendation flips to Claim + use emergency savings : product neutrality, no credit pushed.
Home tab: Turn on alerts	"Your INR 12,000 EMI is due in 2 days, INR 4,600 short", "Unusual INR 8,500 debit", "Failed INR 2,450 payment not refunded"; See my options opens the case.
Untick consent, start a case, then Grant consent	Graph pauses at the consent gate with zero tool calls, then resumes.
Change the language to Tamil and start a case	Explanation in Tamil; "Original (English)" under it; amounts identical.
Press the mic, speak in Hindi	Transcript appears in the message box for review before sending.
Passport tab: Download PDF	One-page Resolution Passport ready to save or print.
Support login (9999): open a Human review case	Pick up, verify documents, recommend an option, message the customer.
Trust ledger tab	Every MCP call with allow or deny, approvals with payload hash, partner callbacks.

07 Demo script (about 8 minutes)

Before every rehearsal: `npm run reset` (so alerts reappear), then `npm run demo`, then open `http://127.0.0.1:8000`.

1. **Opening line:** "Financial apps make you choose the product. Saathi starts with your problem."
2. **Home (30 s):** sign in as Riya, show the Financial Twin headline and the bills inbox going negative before payday. Turn on alerts.
3. **Hospital (2 min):** tick consent, press the RECOVER chip (or speak it). Show the gap tiles and open "How the cover was calculated".
4. **Options (1 min):** compare the five paths, open "Why this score?", point out commission is never an input.
5. **Approve (1 min):** Review & approve: payment link, loan Key Fact Statement, payload hash. Approve and switch to Timeline as partner updates arrive.
6. **Agents and trust (1 min):** Agents tab (nodes, MCP tools, AA consent, playbook allowlist), then Trust ledger.
7. **Ask Saathi (30 s):** "Why is the 3-month plan better than using my savings?" Switch language to Hindi for the reply.
8. **Platform proof (1 min):** Failed UPI refund (pure YAML), then UPI fraud dispute. Same engine, no new code.
9. **Passport and handoff (30 s):** Download the Passport PDF; sign in as support and pick up a case.
10. **Close:** "One conversation, one financial context, one action plan, one timeline." Optional: Arjun's recommendation flip.

08 Trust rules enforced in code

Consent and approval

- Nothing is read without purpose-bound case consent (revocable)
- Writes need a 5-minute approval token bound to case, action, person, and payload hash
- A changed amount or step invalidates the approval
- Each approved write executes at most once
- Voice, AI answers, and OCR each need their own consent

Gateway checks on every tool call

- Tool on the allowlist and declared by the active playbook
- Strict input schema, same case only
- Only the owner (or the assigned specialist, read-only)
- Partners re-check: payment link within what the customer can pay, refund compensation within INR 100/day after T+1, dispute amount equals the debit

AI boundaries

- The model never calculates; the explainer only uses decision numbers
- OpenAI answers are checked for invented numbers and replaced if found
- Translations keep formula lines verbatim and fall back if an amount changes
- Contact details are redacted before any external call

Privacy and audit

- Original uploads and audio are never stored
- Partners see only the tool arguments and the customer ID
- The audit log stores tool, scope, decision, and references, never document contents or keys
- Customers cannot read each other's cases (404)

09 Run it, accounts, and settings

COMMAND	PURPOSE
<code>npm install</code>	Install dependencies (needed after pulling: LangGraph.js, MCP SDK, yaml were added)
<code>npm run dev</code>	Development: web on :5173, API on :8000
<code>npm run reset</code>	Clear demo cases so alerts reappear
<code>npm run demo</code>	Build and serve everything on :8000 with no file watcher (use for presenting)
<code>npm test</code> / <code>npm run typecheck</code>	60 backend tests / type checks for both projects
<code>npx @modelcontextprotocol/inspector</code>	Open a partner server: Streamable HTTP, <code>http://127.0.0.1:8000/mcp/insurer</code> , headers <code>Authorization: Bearer <token></code> and <code>x-saathi-customer: demo-customer-01</code> (token printed at startup or set via <code>MCP_PARTNER_TOKEN</code>)

DEMO ACCOUNT	PASSCODE	USE IT TO SHOW
<code>demo-customer-01</code> (Riya)	2468	Hero journeys, tight savings, EMI and refund cases, alerts
<code>demo-customer-02</code> (Arjun)	1357	Healthy savings, recommendation flips to savings
<code>support-agent-01</code>	9999	Specialist desk

Key `.env` settings: `OPENAI_API_KEY` + `OPENAI_ENABLED=true` , `SARVAM_API_KEY` + `SARVAM_ENABLED=true` , `SARVAM_TRANSLATE_MODEL=sarvam-translate:v1` , `DEMO_DATE=2026-10-03` (keeps the EMI two days away), `MCP_PARTNER_TOKEN` , optional `MCP_URL_<SERVER>` and n8n settings. Never commit `.env` .

10 Limits and what is next

Before demo day: walk through every screen in a real browser, including phone width; the UI has been type-checked and built but not visually reviewed by the developer assistant. Rotate the OpenAI and Sarvam keys after the hackathon (they were shared in a chat).

Honest limits of this prototype

- All partner outcomes (claim, loan, dispute, refund) are simulated
- Synthetic data only; demo passcodes are not secrets
- Policy retrieval is a keyword index (Cognee not connected)
- Lender offers are labelled fixtures (Mochatrade not connected)
- Scanned PDFs are not rendered (photos are OCR'd)
- SQLite and in-process events instead of Postgres and Redis
- Security and consent are prototype design, not certification

Next steps in the plan

- **Cash-flow Copilot:** 7/30-day forecast, shortfall day, one tap into the right playbook
- **"Can I afford it?":** cash vs EMI vs wait scenarios with buffer impact
- **Trust plane upgrades:** consent IDs with 72h expiry, hash-chained audit ledger with a verify button, circuit breaker, response sanitiser
- **Saathi Health Score** and **Emergency mode**
- **Demo-proofing:** golden scenario tests, latency budget display, rehearsal

Paytm Saathi / Team Swamit Labs / Track 2: AI-Powered Financial Journeys. Prepared from the code at commit c25e85d. Synthetic demonstration; no real claim, credit, payment, or partner action is performed.